a **Genealogy cohorts: worlds and nested lines**

Cohort	Worlds	Confined worlds	Lines from confined worlds	Open-first lines
N3-v6	20	17	1088	1088
N6	20	20	1280	1280

64 sampled-cell ancestries / confined world; world = simulation
N3-v6: discovery; N6: confirmation; Z80 tape length L = 16 bytes
Copier: ≥75% tape bytes copied in ≥32/64 uniform partners
Counted trace = fetch of any partner opcode or operand byte
Genealogy fetch class: open ≥32/64 counted traces; confined <32/64
Open-first: earliest copier on each sampled line is classified open

Sampled lines: 64 ancestries per world with a confined founder
Worlds column counts all worlds; genealogy classes use 64 partner traces

b **First founders versus later conversions**

Cohort	First founders, worlds	Later founder events, n	Total founder events
N3-v6	17	15	32
N6	20	14	34

20 worlds/cohort; at most one first confined founder/world
Founder: first confined copier after latest open copier on sampled line
Open ancestry to confined copier; classification as in panel a
Later events follow the first founder in that world; events can repeat
Later events can reuse existing confined-cell byte sources

First founders and later confined-founder events stay separate
Worlds, ancestral lines and founding events are different units

c **N6 first-founder mechanisms**

Recorded mechanism	Founders /20
novel	18/20
damage	2/20

novel = rewrite, <75% byte match to either parent (18)
damage = partial overwrite, ≥75% parent byte match (2)

Separate N6-1 source rule: copy/rewrite 18/20, mutation 0/20
Recorded event categories and causal source classes differ

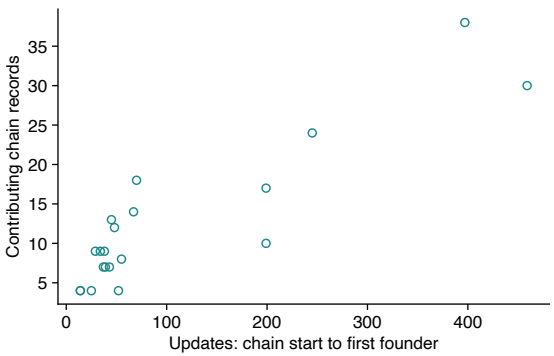
d **N6 founding executors: two copier scopes**

Executors, n	Whole-tape	Executed-byte
Non-copier	20	11
Copier	0	9

Executed-byte: 11/20 non-copiers; requires ≥12/20; not met
Whole-tape: 20/20 non-copiers; requires ≥16/20
Soup-abundance null: P(all 20 whole-tape non-copiers) = 0.062
Null draws executors by pre-founder soup composition

One founding executor/first confined founder: 20 N6 worlds
Copier: ≥75% scoped bytes copied in ≥50% partner encounters
Scopes: entire tape / only executed instruction-stream bytes

e **N6: all first-founder chain lengths**



N6: one point per first founder, 20 worlds
Records = contributing chain events; intervening whole-tape donors are non-copiers (executed-byte function kept separate)

f **Registered N6: complete component ledger**

ID	Requirement	Observation	Verdict
N6-1	Copy/rewrite ≥16/20; mutation ≤2/20 founders	Copy/rewrite 18/20 Mutation 0/20	met
N6-2	Whole-tape non-copier chains ≥15/20	20/20	met
N6-3	Median cell carrier share ≥5× random-tape baseline	Observed 0.00190 Baseline 0.00024 Ratio 7.9×	met
N6-4	Non-copier executors whole-tape ≥16/20; executed-byte ≥12/20	Whole 20/20 Executed-byte 11/20 Null P = 0.062	**not met**
N6-5′	Expected mutation founders <0.05 in 20-world cohort	0.0064 expected 0/20 observed	met

N6-3: cells carrying a word / all soup cells; 2-byte word at executed start
Median across 99 word/founder rows in pre-founder snapshots
Mutation-founded first founders: 0/20; N6 rejection threshold ≥5/20
Input: 55 point-mutation events on 20 pre-founder chains